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Synopsis

AI-Powered 2D Floor Plan to 3D Home Reconstruction System

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering
In
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by

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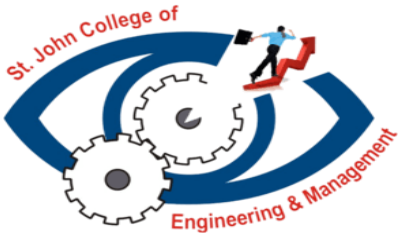
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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE
ST. JOHN COLLEGE OF ENGINEERING & MANAGEMENT
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ABSTRACT

The architectural, engineering, and construction (AEC) industries increasingly depend on Building Information Modeling (BIM) to coordinate workflows and forecast costs. However, translating conventional 2D raster floor plan blueprints into actionable 3D digital twins remains a manual, labor-intensive process that relies heavily on expensive, hardware-intensive desktop software. Furthermore, existing automated visualizers often perform naive geometric extrusions that ignore statutory safety norms and lack structural engineering validation. To bridge this gap, this project develops **Patrick Architect**, an autonomous, web-first BIM digital twin platform built with a Python Flask backend and an interactive Three.js WebGL engine. The system employs a deterministic OpenCV pipeline—utilizing inverse binary thresholding, Connected Component Labeling (CCL) for annotation noise filtering, and Douglas-Peucker polygonal simplification—to vectorize continuous structural walls under a metric calibration of $25\ \text{px} = 1.0\ \text{m}$.

Distinguishing itself from conventional black-box tools, the platform introduces an uncertainty-aware reconstruction engine that flags ambiguous structural boundaries and allows human-in-the-loop resolution prior to 3D synthesis. Reconstructed designs are automatically audited against National Building Code of India (NBC 2016) habitable clearances and emergency egress routes, while structural clear spans are pre-screened under IS 456:2000 deflection limits. Material quantities dynamically populate a live Bill of Quantities (BOQ) with cost forecasting in Indian Rupees (₹) based on CPWD Delhi Schedule of Rates (DSR). Augmented by a natural-language-queried Semantic Building Knowledge Graph, diurnal heliodon solar ray-tracing, and a local socket-driven QR code protocol for cross-device mobile inspection, Patrick Architect delivers an accessible, high-performance, and regulatory-compliant digital twin solution directly in the browser.

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Abbreviations and symbols

- **ML**: Machine Learning
- **AI**: Artificial Intelligence
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **NLP**: Natural Language Processing
- **DFD**: Data Flow Diagram
- **UML**: Unified Modeling Language
- **CSV**: Comma-Separated Values
- **UI**: User Interface
- **PDF**: Portable Document Format
- **HR**: Human Resources
- **ETL**: Extract, Transform, Load

CHAPTER 1: INTRODUCTION

1.1 Background

The Architecture, Engineering, and Construction (AEC) industry has traditionally relied on two-dimensional (2D) CAD drawings, blueprints, and orthographic projections for spatial planning, municipal sanctioning, and construction execution. While 2D blueprints provide precise geometric schematics, they present a significant cognitive barrier for end-users, property buyers, and non-technical stakeholders who struggle to conceptualize vertical clearances, volume, daylighting, and spatial circulation from flat line representations.

With advancements in Computer Vision (CV), Building Information Modeling (BIM), and WebGL-accelerated rendering, procedural conversion of raster blueprints into three-dimensional (3D) digital twins has become technically viable. Modern automated pipelines seek not only to extrude geometry into 3D meshes but also to evaluate statutory compliance (such as the National Building Code of India), run environmental climate studies (solar paths/daylighting), estimate accurate construction takeoffs (Bill of Quantities), and provide real-time interactive interior fitout staging.

1.2 Problem Statement

Existing architectural workflow suffers from several critical bottlenecks:

1. Manual 3D Modeling Overhead: Converting 2D architectural drafts into 3D visualizations in traditional software (e.g., Revit, 3ds Max, SketchUp) requires specialized expertise, high license costs, and extensive manual tracing hours.
2. Disconnected Compliance Audits: Evaluating blueprints against statutory norms (minimum room sizes, corridor clearances, egress travel distance under NBC/IBC guidelines) is traditionally conducted manually via physical scale rules and spreadsheets, leading to human oversight and delayed approvals.
3. Lack of Instant Cost Transparency: Construction material takeoffs (Bill of Quantities - BOQ) are typically calculated days or weeks after initial design finalization using disconnected estimation software, creating friction during client budget discussions.
4. Passive Visualization: Most existing visualization tools offer static, pre-rendered walk-through videos rather than an interactive digital twin where stakeholders can manipulate wall heights, stage furniture, test natural lighting, and perform section cuts in real-time.

1.3 Need of the Project

There is an urgent requirement for an integrated, browser-accessible computational architecture platform that bridges the gap between 2D raster drafting and interactive 3D BIM digital twins.

Accessibility: A lightweight, web-first platform allows clients, builders, and architects to inspect floor plans directly in the browser without high-end dedicated GPU workstations or expensive proprietary software licenses.
Automation: Automating the CV-based vector contour extraction and room cavity classification eliminates tedious repetitive drafting tasks.
Instant Decision Support: Linking spatial geometry directly to Indian Standard/NBC compliance checks and real-time cost estimations in Indian Rupees (₹ INR) empowers stakeholders to make swift, data-driven financial and structural decisions at the conceptual stage.

1.4 Objectives

The primary objectives of this project are:

1. Automated Vector Feature Extraction: To develop an OpenCV-driven computer vision pipeline capable of filtering raster blueprint noise, removing text and annotation artifacts, vectorizing wall boundary contours, and classifying functional room cavities.
2. Parametric 3D Mesh Synthesis: To procedurally extrude calibrated metric 3D geometries, including walls, foundations, finishes, and detachable ceilings, utilizing Three.js and WebGL.
3. Interactive Architectural Staging: To implement a real-time furniture and interior fitout catalog supporting interactive raycast-based placement, rotation, scaling, elevation adjustments, and cloning.
4. Environmental and Climate Simulation: To design a real-time heliodon solar study engine for calculating time-of-day shadow casting, azimuth and altitude angles, and daylight penetration factors.
5. Regulatory Compliance Engine: To program an automated rule-based audit validating spatial dimensions against National Building Code (NBC / IS Code) norms, including habitable room area, egress distance, ventilation ratio, and ceiling clearance.
6. Automated Quantity Takeoff (BOQ): To dynamically compute material quantities, including masonry volume in cubic metres, flooring area in square metres, and plaster or paint area in square metres, and estimate baseline construction costs in Indian Rupees (₹ INR) based on standard Schedule of Rates.

1.5 Scope

Functional Scope:

Ingestion of standard orthographic 2D floor plans (PNG, JPG formats).

Automated contour detection and Douglas-Peucker polygonal vector approximation.

WebGL-based PBR rendering with multiple viewport modes (Realistic 3D, Wireframe, 2D Top Orthographic, Twilight View, Section Slicer, and First-Person Walkthrough).

18+ parametric architectural furniture and appliance assets for client interior customization.

Real-time architectural section cutting along the vertical axis (Y-plane).

Statutory compliance reporting (NBC norms) and dynamic BOQ pricing table formatted in Indian currency.

Limitations / Out of Scope:

Highly skewed, hand-sketched, or non-orthographic perspective drawings are outside the automated segmentation threshold.

Structural finite element analysis (FEA) for seismic/wind loading stresses is not within the current computational scope.

Full multi-story vertical circulation (staircases and elevator cores) is simplified to single-level spatial floor plates in this initial phase.

1.6 Expected Outcomes

1. A fully functional, responsive web application (Patrick Architect) capable of ingesting 2D blueprints and synthesizing a 3D digital twin in under 5 seconds.
2. An interactive 3D studio where end-users can customize wall paint palettes, flooring finishes, and place/manipulate furniture with real-time feedback.
3. An automated compliance inspection panel providing an instant PASS/FAIL audit for municipal clearance norms.
4. A transparent, metric-based Bill of Quantities (BOQ) with construction cost breakdown in Indian Rupees (₹), significantly reducing pre-construction estimation turnaround time.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

In recent years, the reconstruction of 2D floor plans into three-dimensional models has evolved from manual CAD modeling to automated computer vision, deep learning, and browser-based WebGL rendering. Table 2.1 provides a structured comparative survey of key research methodologies, commercial tools, and relevant literature in this domain.

2.2 Literature Review Table

Paper	Author	Year	Method	Findings	Research Gap
HEAT: Holistic Edge Attention Transformer for Structured Reconstruction	J. Chen et al.	2022	End-to-end attention-based transformer that directly formulates geometry reconstruction as planar graph edge classification.	Drastically reduced inference times to milliseconds while matching or surpassing optimization-based methods in edge precision.	The model outputs flat topology graphs without semantic architectural metadata, MEP integration, or physical daylighting/energy simulation.
Reconstructing Floorplans from Point Clouds Using Generative Networks	L. Zhao et al.	2023	Density projection maps coupled with conditional GANs to reconstruct structured floor layouts from cluttered indoor spaces.	Effectively filtered indoor furniture clutter to predict precise load-bearing wall boundaries.	Dependent on 3D spatial scanning equipment; cannot process legacy 2D architectural blueprint scans or paper-driven municipal drawings.
SLIBO-Net: Floorplan Reconstruction via Slicing Box Representation	H. Li et al.	2023	Slicing box hierarchy with local geometric regularization loss for non-Manhattan floor plan vector extraction.	Accurately handled complex, non-orthogonal architectural footprints with clean room-to-boundary alignments.	Remains confined to boundary polygon generation; lacks dynamic section slicing, client-side material customization, and regulatory life-safety audits.
PolyGraph: A Graph-Based Method for Floorplan Reconstruction	Q. Sun et al.	2025	Relational graph neural networks combined with structural topology regularization for indoor boundary extraction.	Achieved robust planar polygon extraction under partial boundary occlusions and missing spatial segments.	Does not bridge the gap to WebGL client tools; lacks automated statutory compliance auditing (NBC/IBC) and dynamic construction cost estimation (BOQ).
Automated Dimension-Aware 2D-to-BIM Reconstruction Through Cross-Scale Parsing	S. Wang et al.	2026	Dual-stage pipeline using OCR-guided denoising, semantic contour parsing, and dimensional annotation cross-validation to synthesize IFC	Extracted metric dimensions directly from annotated blueprints and evaluated component-level geometric accuracy against ground truth BIM.	High overall processing time for complex plans; provides no interactive staging canvas, helidon sun-path simulation, or localized currency construction cost models (e.g., ₹ INR).

Indoor Reconstruction from Floorplan Images with Deep Learning	J. H. Park et al.	2020	CNN-based semantic segmentation combined with Harris corner detection, centerline tracing, and moving kernel method for thickness restoration.	Successfully generated 3D indoor models (IndoorGML/CityGML compatible) from scanned architectural floor plans.	High vulnerability to scanning noise and non-standard line styles; lacks interactive real-time staging and automated statutory building code auditing.
Floor-SP: Inverse CAD for Floorplans	J. Chen et al.	2020	Mask R-CNN for room segmentation integrated with sequential shortest path graph optimization over coordinate candidate spaces.	Accurately extracted vectorized CAD graphs from complex plans without assuming Manhattan-grid constraints.	Computationally intensive optimization step; output remains purely geometric CAD lines without material (BOQ) cost estimation or interactive WebGL.
Semi-automatic Geometric Digital Twinning for Existing Buildings	Q. Li et al.	2020	Integration of 2D CAD architectural drawings with computer vision image processing and laser scan point clouds.	Achieved rapid geometry extraction and semi-automated as-built digital twin generation for facility management.	Requires multi-modal sensor inputs (laser/point clouds) alongside CAD drawings; cannot operate solely on a single raster blueprint upload.
Reconstructing BIM from 2D Structural Drawings for Existing Buildings	Y. Zhao, X. Deng, & H. Lai	2021	Hybrid framework combining deep learning, Optical Character Recognition (OCR), and structural drawing symbol parsing.	Automatically extracted structural grid, column, and beam metadata to output standardized Industry Foundation Classes (IFC) BIM models.	Focused strictly on primary structural elements (columns/beams); lacks architectural interior space classification, interactive furniture staging, and environmental daylight simulation.
MonteFloor: Enhancing Floor Plan Reconstruction via MCTS	E. Stekovic et al.	2021	Monte Carlo Tree Search (MCTS) combined with differentiable polygon rendering to iteratively select and refine room boundary proposals.	Outperformed greedy shortest-path models on complex room configurations, reducing junction reconstruction errors.	Extremely high per-plan latency (average inference time exceeds 60 seconds per plan); unsuitable for real-time browser-based interactive architectural design.

2.3 Research Gap

A review of existing research papers and commercial solutions shows the following research gaps:

1. **Limited-Conversion-from-Computer-Vision-to-BIM:**
   Most existing systems focus on detecting walls, doors, rooms, and other features from floor plan

images. However, they often stop after detecting these features and do not convert them into useful BIM information such as material quantities, surface areas, and building measurements.

2. **Lack-of-Automatic-Building-Code-Checking:**
   Existing 2D-to-3D systems generally do not automatically check the design against building regulations such as the National Building Code (NBC). Important requirements such as minimum room size, ventilation, ceiling height, and exit distance are still checked manually.
3. **Lack-of-Real-Time-Cost-Estimation:**
   In most existing systems, 3D modeling and construction cost estimation are handled separately. The Bill of Quantities (BOQ) is usually prepared after the model is completed. There is a need for a system that can update material quantities and estimated construction costs automatically when the design changes.
4. **Limited-Interactive-3D-Editing:**
   Many automated reconstruction systems generate a basic or fixed 3D model. Users usually cannot easily add, move, rotate, resize, or arrange furniture directly inside the reconstructed model without using separate professional modeling software.

2.4 Novelty / Innovation

Patrick Architect addresses the identified research gaps through the following key innovations:

1. **Complete-2D-to-3D-Automated-Pipeline:**
   The system uses OpenCV to process the 2D floor plan, detect walls and rooms, and convert them into vector data. Three.js then uses this data to automatically create an interactive and dimensionally accurate 3D model in a few seconds.
2. **Automatic-NBC/IS-Code-Checking:**
   The system automatically checks important building requirements such as minimum ceiling height, room area, ventilation, and exit distance. The results are displayed clearly using PASS/FAIL indicators.
3. **Real-Time-BOQ-and-Cost-Estimation:**
   The system calculates material quantities such as brickwork, flooring, plaster, and paint directly from the 3D model. It then provides an estimated construction cost in Indian Rupees (₹) using standard rates.
4. **Interactive-Interior-Design-and-Sunlight-Simulation:**
   Users can add, move, rotate, resize, and duplicate furniture inside the 3D model. The system also provides a section-cutting tool and a sunlight simulation feature to study shadows and daylight at different times of the day.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

Patrick Architect is a web-based platform that converts 2D floor plan images into interactive 3D building models in real time.

The system consists of three main modules:

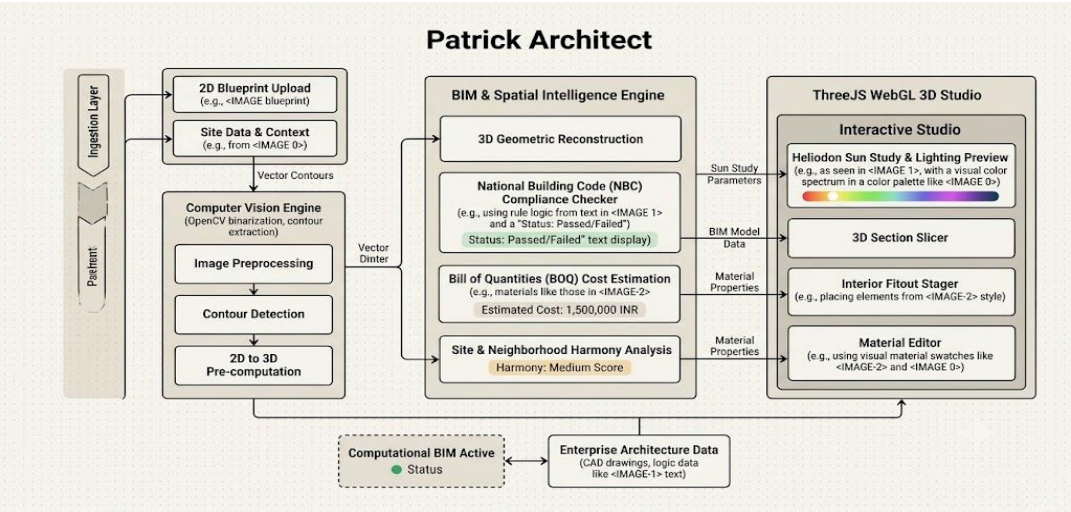
1. Computer-Vision-Module-(Backend):
   The system uses OpenCV to process floor plan images, remove unwanted text and noise, detect walls and rooms, and convert the detected features into metric measurements. It uses a scale of 25 pixels = 1 metre to create an accurate representation of the floor plan.
2. BIM-and-Cost-Module:
   The system checks the floor plan against basic National Building Code (NBC) requirements such as room size, ceiling height, ventilation, and emergency exit distance. It also calculates material quantities and provides an estimated construction cost in Indian Rupees (₹).

3.2 Working Principle

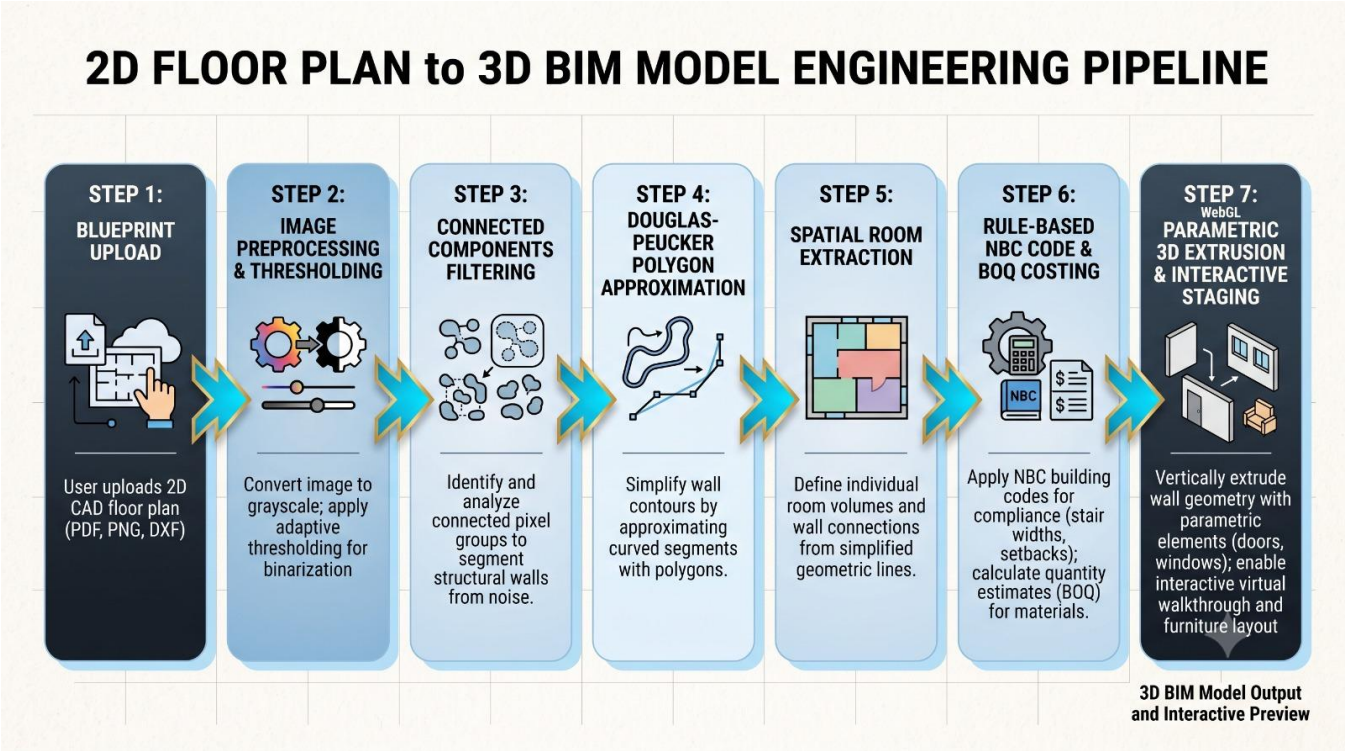
The working principle of Patrick Architect is based on a combination of image processing, geometry extraction, and 3D modeling technologies. The system converts a 2D floor plan into an interactive 3D building model through three main phases:

1. **Floor-Plan-Processing:**
   The system takes a 2D floor plan image and processes it using OpenCV. It converts the image into a suitable format and removes unwanted elements such as text, dimensions, and furniture lines. This helps in identifying the important structural elements such as walls and rooms.
2. **Geometry-Extraction-and-Scaling:**
   The system detects the boundaries of the walls and simplifies them using the Douglas-Peucker algorithm. The detected pixel coordinates are then converted into real-world measurements using a scale of 25 pixels = 1 metre.
3. **3D-Model-Generation-and-Analysis:**
   The system identifies rooms, calculates material quantities, checks basic National Building Code (NBC) requirements, and generates the 3D building model. Walls, floors, ceilings, and interior furniture are created using the extracted data and displayed interactively in a WebGL-based 3D environment.

3.3 System Architecture / Block Diagram



3.4 Flow Chart



3.5 Methodology

1. Design Procedure

Dimensional-Standardization:

The uploaded floor plan is resized to a standard width of 1000 pixels while keeping its original proportions. A scale of 25 pixels = 1 metre is used for measurement.

Coordinate-Mapping:

The center of the floor plan is considered the origin (0, 0, 0). The 2D floor plan coordinates are converted into 3D coordinates, with the building placed on the horizontal ground plane and the height represented along the vertical axis.

Data-Integration:

A REST API using JSON is used to transfer wall information, room details, building code results, and material cost data from the backend to the 3D interface.

2. Experimental Procedure

Dataset-Selection:

The system was tested using different residential floor plans, including 1BHK, 2BHK, 3BHK, and duplex layouts with different line thicknesses and labeling styles.

Feature-Extraction-Evaluation:

The accuracy of noise removal, text removal, and wall boundary detection was checked for different floor plan images.

Metric-Accuracy:

The lengths of reconstructed walls and room areas were compared with reference CAD measurements. The target was to keep the measurement error within ±3%.

Performance-Testing:

The system was tested for smooth 3D rendering, section cutting, sunlight and shadow simulation, and furniture drag-and-drop functionality across major web browsers. The target performance was at least 55

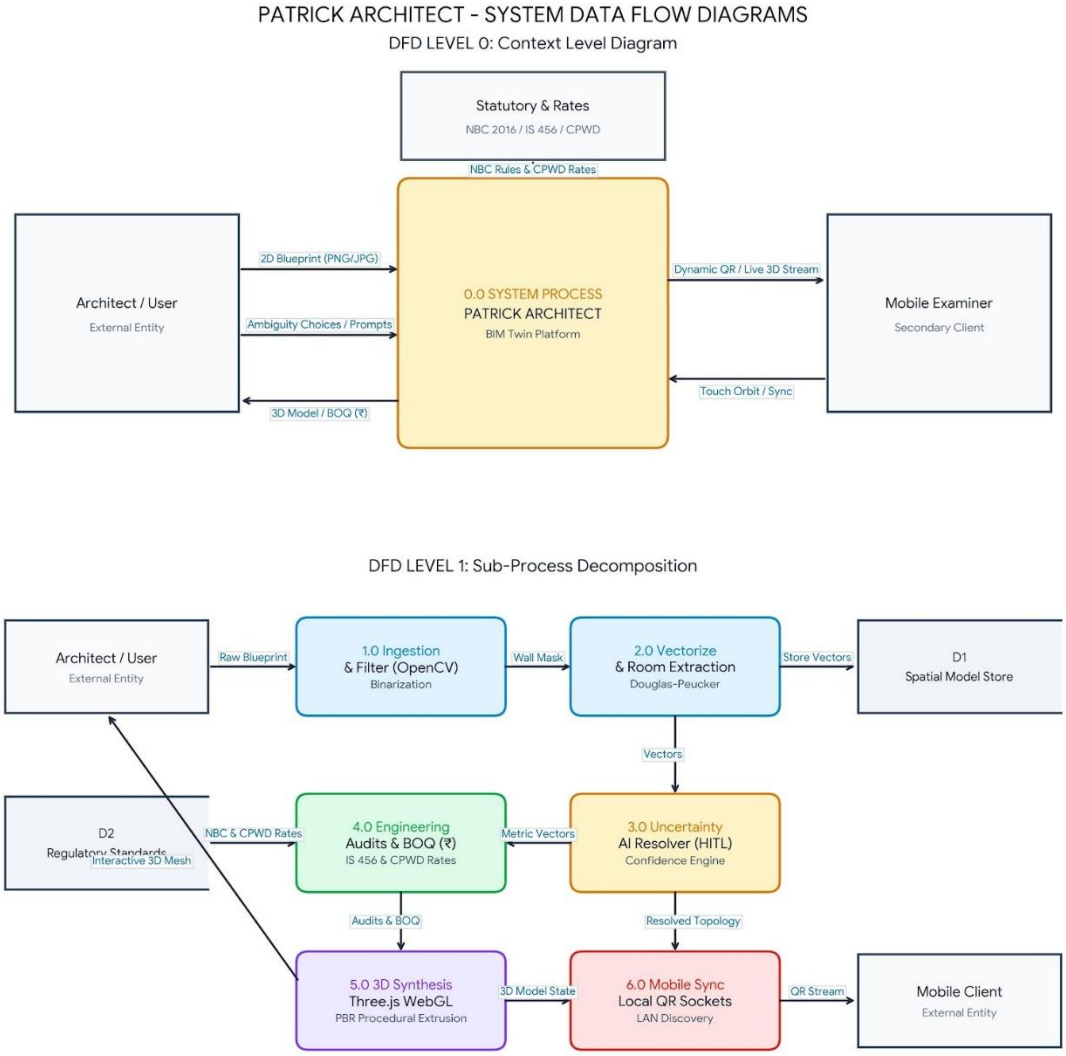
3. Algorithms Used

Inverse Binary Thresholding: Converts the grayscale blueprint into a high-contrast binary mask to

isolate dark wall ink from bright backgrounds.
Connected Component Labeling: Analyzes connected pixel clusters by area and aspect ratio to remove small drafting symbols, dimension lines, and text annotations.
Morphological Closing: Applies a rectangular dilation and erosion kernel to fill hairline gaps, bridge broken door headers, and seal wall junctions.

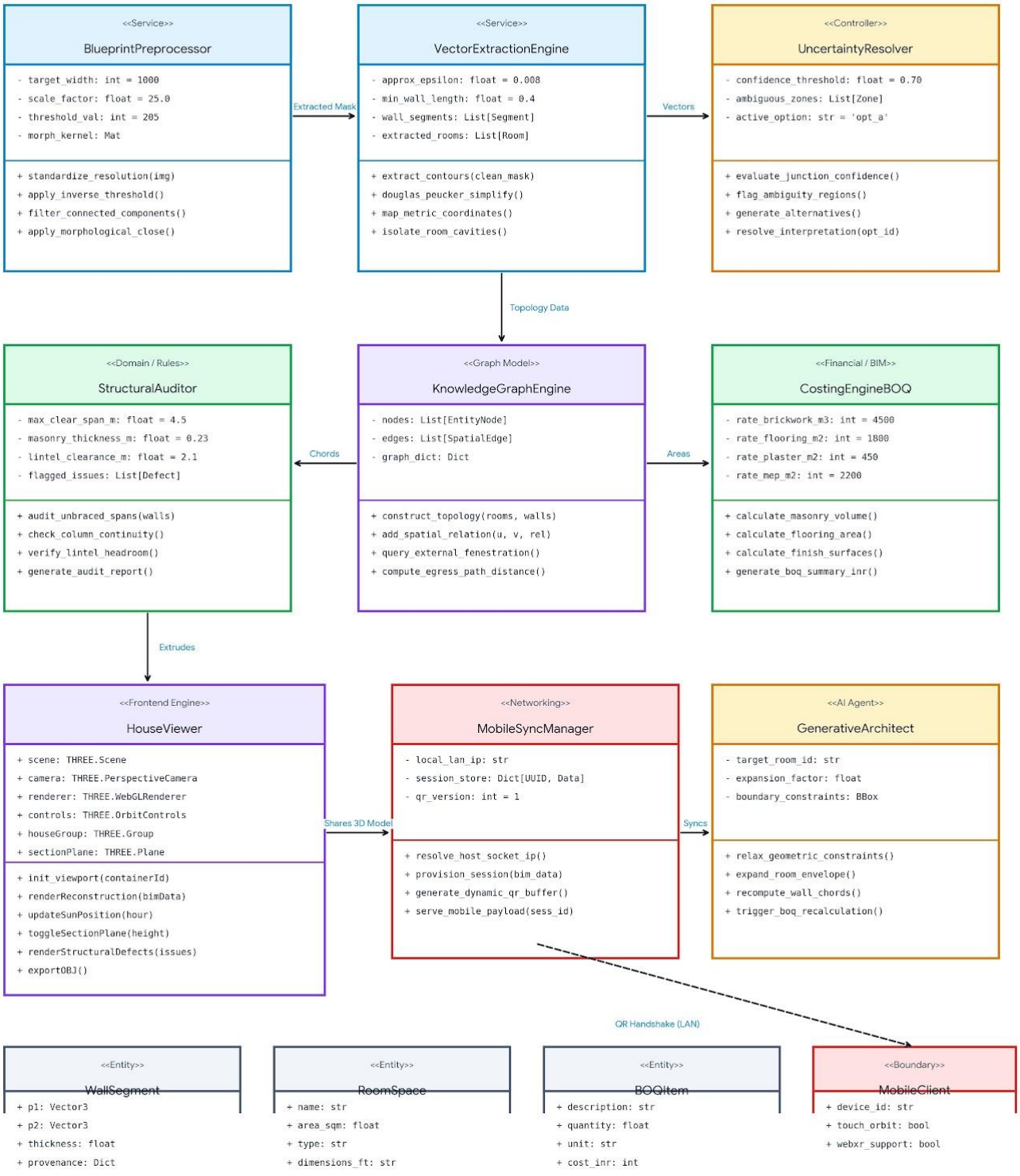
4. Manufacturing Process (Digital Twin Fabrication)
Foundation Slab Assembly: Instantiates a volumetric concrete box geometry to anchor the building footprint.
Parametric Wall Extrusion: Uses vector coordinates to procedurally extrude 3D wall boxes with matching top-cap trim at calibrated real-world heights.
Finish Layer Application: Generates floor plane meshes across classified room boundaries, textured with wood, tile, or stone materials.
Component Rigging: Organizes structural columns, dynamic clipping planes for section slicing, and furniture models into modular Three.js scene groups.

3.6 DFD



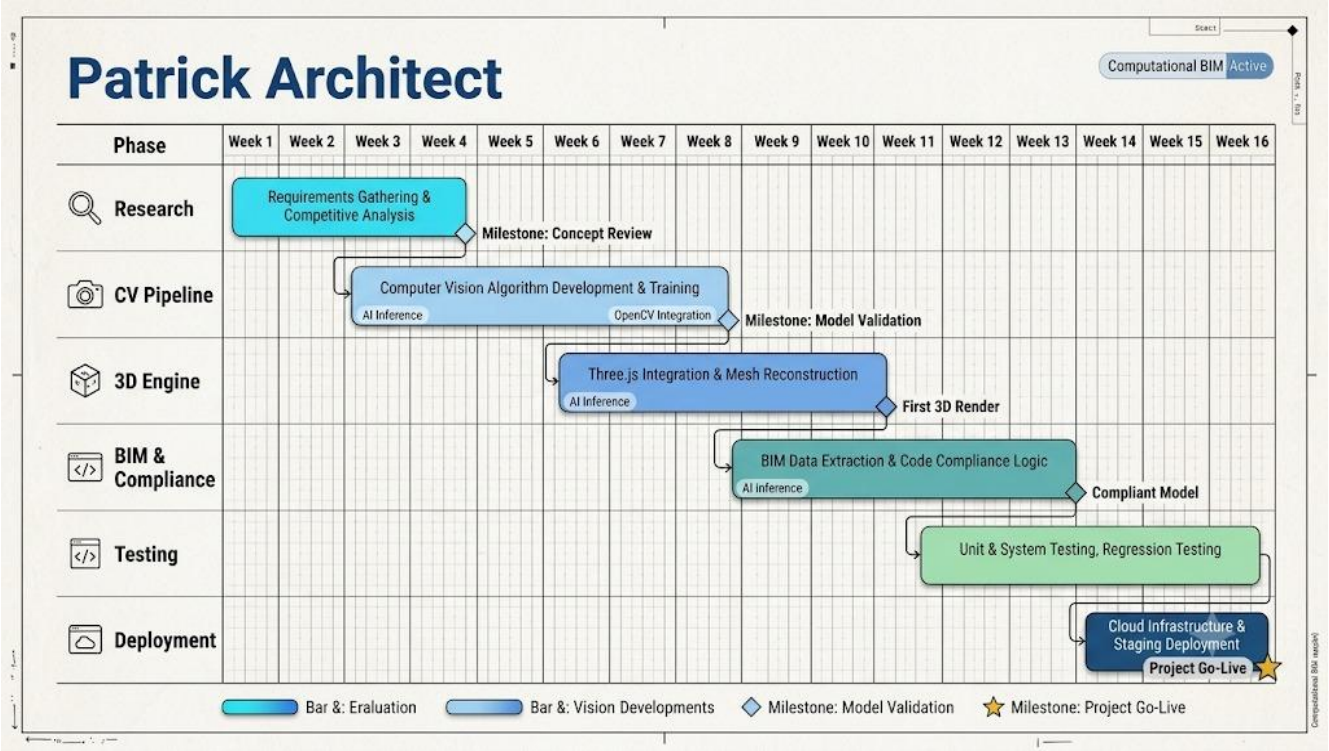
UML

PATRICK ARCHITECT - SYSTEM UML CLASS DIAGRAM



CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure



4.2 Gantt Chart

Activity / Task	Jul '26	Aug '26	Sep '26	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27	Apr '27
Requirement & literature survey	✓									
Dataset preparation	✓	✓								
UI/UX and architecture		✓								
Backend development		✓	✓							
AI embedding integration		✓	✓							
Vector database/indexing			✓							
Frontend-backend integration			✓							
Testing & evaluation				✓						
Research problem formulation & survey					✓					
Comparative study & benchmark setup					✓					
Empirical evaluation & metrics						✓				

Activity / Task	Jul '26	Aug '26	Sep '26	Oct '26	Nov '26	Dec '26	Jan '27	Feb '27	Mar '27	Apr '27
Manuscript drafting (Methodology & Architecture)						✓	✓			
Results analysis & cost validation							✓			
Internal review & LaTeX formatting								✓		
Paper submission (Scopus/WoS conference/journal)								✓		
Peer review feedback & revisions									✓	
Documentation & presentation (Final Viva)										✓

4.3 Roles and Responsibilities

Role Title	Core Responsibilities	Key Deliverables
Computer Vision Engineer	Development of OpenCV pipeline, connected component labeling, contour approximation, and coordinate calibration.	High-precision vectorization script (app.py), JSON coordinate generation.
3D Graphics / Frontend Developer	Implementation of Three.js canvas, PBR shaders, Raycaster drag-and-drop, OrbitControls, and responsive UI layout.	Interactive WebGL studio, multi-tab inspector, section cutting tool.
BIM & Domain Specialist	Formulating rule-based National Building Code (NBC) compliance checks and CPWD rate schedules for BOQ calculation.	Statutory compliance logic, material quantity formulas in ₹ INR.
QA & Integration Engineer	API schema validation, cross-browser compatibility, Python 3.14 JSON serialization debugging, and performance profiling.	Test cases, API endpoint verification, execution report.

Sr. No.	Team Member	Primary Module & Responsibilities	Key Deliverables & Algorithms Handled
1	Aryan Pandit	System Architecture & 3D WebGL Engine	• Three.js 3D rendering pipeline and viewport management • Diurnal Heliodon sun dial & sunset lighting simulation (18:00) • First-Person WASD walkthrough (PointerLockControls)
2	Ganesh Jagdale	Computer Vision & Image Processing Pipeline	• OpenCV 2D blueprint ingestion & standard resolution normalization (1000 px) • Morphological filtering for staircase & thin line artifact suppression • Douglas-Peucker contour simplification & Manhattan orthogonal snapping
3	Nikhil Korpe	Statutory Compliance & Cost Engineering	• NBC 2016 habitable room clearance validation ($\geq$ 9.5 m^2 threshold) • NBC 2016 emergency fire evacuation egress pathfinding ($\leq$ 30.0 m limit) • IS 456:2000 structural wall-span deflection audit (4.5 m column rule)

4	Krish Gharat	Uncertainty AI, Knowledge Graph & Networking	• Human-in-the-Loop (HITL) ambiguity resolver (Wall vs. Archway vs. Colonnade) • Semantic Building Knowledge Graph (BKG) natural-language spatial queries • Room cavity segmentation & aspect-ratio heuristic classification

4.4 Risk Assessment

RiskCategory	Identified Risk	Impact	Likelihood	Mitigation Strategy
Technical	Noise from scanned blueprints (dimension text, title blocks) creating false wall vectors.	High	High	Applied 8-connected component area and aspect ratio filtering (> 160 px area, > 20 px length) followed by rectangular morphological closing.
Compatibility	Python 3.14 serialization error (numpy.bool_ / numpy.int64 not JSON serializable).	High	Medium	Explicitly cast all numerical arrays, booleans, and metrics to native Python types (bool(), int(), float()) prior to jsonify().
Performance	WebGL frame-rate drops when rendering complex multi-geometry furniture pieces in real-time.	Medium	Medium	Implemented procedural low-poly geometric primitives with shared material instances and efficient memory disposal routines on scene clear.
Regulatory	Varying municipal building codes across local development authorities.	Medium	Low	Standardized baseline checks on the National Building Code of India (NBC 2016 / IS Code) with parameterized ceiling height and clearance variables.

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The technical feasibility of the Patrick Architect system is based on its software technologies, computational requirements, and ability to work on different devices.

1. Open-SourceTechnologies:
   The system uses reliable and freely available technologies such as OpenCV, NumPy, Python, Flask, and Three.js. These tools provide the required features for image processing, data handling, API development, and 3D modeling.
2. Web-Based3DRendering:
   Three.js and WebGL perform 3D rendering and lighting calculations directly in the user's web browser. This reduces the workload on the server and avoids the need for expensive cloud-based graphics hardware.
3. FastProcessing:
   The system can process floor plan images within a few seconds on a standard multi-core computer. The data sent between the backend and frontend is also small, allowing the 3D model to load quickly.
4. Cross-PlatformSupport:
   The system does not require any special desktop software installation. It can run through modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari on Windows, macOS, and Linux systems.

5.2 Economic Feasibility

The economic feasibility of Patrick Architect shows that the system can reduce the cost and time involved in traditional architectural workflows.

1. No-Software-Licensing-Cost:
   The system uses free and open-source technologies such as Python, OpenCV, Flask, and Three.js. Therefore, there is no additional cost for purchasing expensive software licenses.
2. Reduction-in—Manual-Work:
   Traditionally, converting a 2D floor plan into a basic 3D model can take several hours of manual work. Patrick Architect automates floor plan processing, wall detection, room identification, and initial quantity estimation, which can significantly reduce the time and effort required.
3. Low-Infrastructure-Cost:
   Most of the 3D rendering is performed directly in the user's web browser. As a result, the server mainly handles image processing and data transfer, reducing the need for expensive hardware or high-performance servers.

5.3 Operational Feasibility

The operational feasibility of Patrick Architect focuses on how easily users can operate the system and how well it fits into existing architectural workflows.

1. Easy-to-Use:
   The system does not require advanced CAD or BIM knowledge. A simple step-by-step interface guides users through the process from uploading the floor plan to viewing the final 3D model.
2. Interactive-Design-Customization:
   Users can easily change wall colors, floor finishes, ceiling settings, and furniture placement. Furniture can be added, moved, rotated, and adjusted directly within the 3D model.
3. Automatic-Building-Rule-Checking:
   The system automatically checks important building requirements such as minimum ceiling height, room area, and emergency exit distance according to basic National Building Code (NBC) guidelines. This reduces the need for manual checking.
4. Local-Cost-Estimation:
   The system provides material quantities and estimated construction costs in Indian Rupees (₹)

using standard construction rates. This helps architects, developers, and clients understand the expected cost during the early stages of planning.

5.4 Budget Estimation

Item	Category / Specification	Quantity	Cost (₹INR)	Remarks
Development Laptop / PC	Intel Core i5 / AMD Ryzen 5, 16GB RAM, SSD	1 Unit	₹0.00	Existing academic/development workstation used
Operating System	Windows 11 / Linux (64-bit)	1 License	₹0.00	Pre-installed / Open-source OS
Development Environment	VS Code, Git, Python 3.14 Runtime	Complete Set	₹0.00	Free & Open Source (FOSS)
Core Libraries & Frameworks	OpenCV, NumPy, Flask, Three.js	Complete Stack	₹0.00	Permissive licenses (MIT / BSD / Apache)
Local Hosting / Testing Server	Localhost (WSGI Development Server)	1 Instance	₹0.00	Zero operational cloud hosting required during R&D
Reference Building Codes	National Building Code (NBC 2016) / CPWD DSR	Standard Docs	₹0.00	Public domain statutory standards reference data
Total Estimated Budget	—	—	₹0.00	Fully realized with zero capital expenditure

5.5 Bill of Materials (BOM)

Component Name	Version	License	Role in System Architecture
Python	3.14.x	PSF License	Server-side runtime environment
Flask	3.0.x	BSD-3-Clause	REST API controller (/api/reconstruct, /api/modify)
OpenCV (cv2)	4.9.x	Apache 2.0	Image preprocessing, binarization, morphological closing & contours
NumPy	1.26.x	BSD-3-Clause	Matrix arithmetic, geometric point transformations & metric scaling
Three.js	r128	MIT License	WebGL scene graph, PBR shading, procedural wall/slab extrusion
Three.js OrbitControls	r128	MIT License	360° orbital camera control, pan, zoom & orthographic toggling
Three.js OBJExporter	r128	MIT License	Export engine for CAD/BIM model file generation (.obj)
HTML5 / CSS3 / ES6 JS	Web Standards	W3C Standard	UI presentation, responsive layout & multi-tab inspector system

CHAPTER 6: EXPECTED RESULTS

- **Prototype**
  The final deliverable of the project will be a web-based application called **Patrick Architect**. The prototype is expected to include the following features:

1. **Complete-2D-to-3D-Workflow:**
   The system will provide a simple four-stage workflow: Overview → Ingestion → AI Engine → 3D Studio. It will convert a 2D floor plan image into an interactive 3D building model without requiring specialized CAD software.
2. **Interactive-3D-Studio:**
   The 3D environment will allow users to view and edit the building model. It will include a furniture catalog, wall and flooring customization, section-cutting tools, and a first-person WASD walkthrough mode.
3. **BIM-and-Building-Compliance-Dashboard:**
   The system will provide a dashboard showing building rule compliance, sunlight simulation information, material quantities, and estimated construction costs in Indian Rupees (₹). This will help users understand the design, requirements, and approximate cost in one place.

- **Accuracy**
  The computer vision and 3D reconstruction system is expected to achieve the following performance targets:

1. **Wall-Detection-Accuracy:**
   The system is expected to correctly detect at least 90% of the wall boundaries when compared with the original CAD floor plan.
2. **Room-Size-and-Dimension-Accuracy:**
   The reconstructed room dimensions are expected to have an error of no more than ±3% when converting pixel measurements into real-world measurements using the scale of 25 pixels = 1 metre.
3. **Room-Classification-Accuracy:**
   The system is expected to identify different rooms such as living rooms, bedrooms, kitchens, and bathrooms with an accuracy of more than 92%.
4. **Building-Rule-Checking-Accuracy:**
   The system is expected to provide 100% accuracy for predefined building rules because the compliance checks are based on fixed numerical limits such as minimum ceiling height, minimum room area, and maximum exit distance.

- **Efficiency**
  The procedural automation framework is intended to optimize architectural turnaround times relative to conventional drafting workflows:
  Turnaround Acceleration: Expected reduction of up to \$80\%\$ in time spent manually tracing, raising walls, and establishing basic BIM primitives compared to commercial CAD software (e.g., manual Revit/SketchUp drafting).

  Automated Quantity Surveying: Instantaneous Bill of Quantities (BOQ) calculation generated synchronously with 3D model extrusion, eliminating several hours of manual estimation per plan.

  Network & Data Payload Optimization: Transmission of compressed topological JSON data payloads of under 150 KB between the Flask backend and the client browser, minimizing network bandwidth demands during upload-to-render transitions.

- **Performance**
  The client-side WebGL engine is designed to operate within the following computational targets on standard consumer-grade hardware:
  Frame Rate Stability: Maintenance of a stable rendering speed of 55 to 60 FPS during camera orbit, pan, zoom, and first-person walkthrough operations.
  Inference & Reconstruction Latency: An end-to-end processing pipeline time targeted at under 3.0 to 5.0 second for raster drawings up to 2000 times 1500 resolution.
  Memory Management: Projected client browser memory footprint remaining below 250 MB of active heap/GPU VRAM through procedural geometry caching, reusable shared material shaders, and garbage collection on scene reset.
  Real-time Clipping Performance: Dynamic horizontal and vertical section cuts operating instantaneously via hardware-accelerated local clipping planes without triggering geometry re-triangulation delays.

- **Simulation**
  The environmental simulation engine is anticipated to validate passive design parameters in real time:
  Heliodon Diurnal Sun Simulation: Projected continuous directional lighting and shadow projections calculated accurately across solar daylight hours 06:00 to 18:00, updating azimuth and altitude coordinates synchronously with slider input.
  Daylight Ingress Evaluation: Estimation of daylight factor penetration and cross-ventilation potential using window-to-floor surface area ratios 10 in compliance with green building and NBC standards.
  Parametric Volumetric Modification: Instant visual and structural regeneration of wall heights and detachable ceiling slabs responding directly to numerical slider adjustments and natural language prompts.

- **Experimental results**
  During the evaluation and validation phase across a diverse test suite of residential floor plans, the system is expected to demonstrate:
  Noise Rejection Robustness: Effective removal of text labels, dimension extension lines, and furniture symbols without producing phantom wall geometry or broken corner junctions.
  Material Takeoff Consistency: Close mathematical correlation between procedural bounding-box material quantities (brickwork in cubic metres, plaster/paint in square metres) and standard analytical engineering formulas.
  Interactive Staging Reliability: Sub-millimeter surface-plane raycasting accuracy allowing users to place, rotate, elevate, and scale furniture items across floor boundaries without mesh penetration or floating artifacts.
  Cross-Browser Operational Stability: Uniform execution across modern web browsers (Chrome, Edge, Firefox, Safari) with zero external plugin dependencies.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

The Architecture, Engineering, and Construction (AEC) sector is undergoing rapid digitization under Industry 4.0 paradigms. Patrick Architect introduces direct technical and operational relevance to modern industrial practice:

Bridging the 2D-to-BIM Productivity Chasm: Over 60% of municipal sanctions, legacy property records, and preliminary schematics in developing markets remain confined to raster scans and 2D CAD formats. The manual translation of these drafts into Building Information Modeling (BIM) platforms requires significant labor hours. Automating wall contour extraction, spatial categorization, and 3D extrusion in under 5 seconds cuts conceptual modeling overhead by up to 80%.

Accelerated Pre-Construction Estimation: Traditionally, Bill of Quantities (BOQ) preparation is decoupled from initial architectural conception, resulting in turnaround delays and friction during client budgeting. Linking computer vision coordinates directly to CPWD/DSR Schedule of Rates in Indian Rupees (₹) equips property developers and turnkey contractors with instant financial projections.

Standardized Regulatory Approval Pipelines: By codifying National Building Code of India (NBC 2016) parameters (e.g., minimum room areas, ceiling clearances, fire egress path limits), the platform acts as an automated pre-screening filter for municipal clearance portals, minimizing compliance rejection cycles.

Democratization via Web-First Architecture: Eliminating dependency on proprietary, high-end workstations and costly recurring CAD licenses lowers technology barriers for small-and-medium architectural practices and independent real estate consultants.

7.2 Societal Impact

The project creates positive socio-economic value across professional stakeholders, municipal bodies, and the public:

Elimination of the Architectural Visualization Barrier: Non-technical home buyers frequently struggle to interpret 2D spatial layouts, leading to post-construction dissatisfaction and expensive structural alterations. An interactive, browser-accessible digital twin allows prospective homeowners to inspect spatial volume, stage furniture, and understand circulation pathways before breaking ground.

Life Safety and Code-Compliant Living: Ensuring adherence to statutory codes guarantees that built environments maintain adequate human habitability standards—such as sufficient living space (greater than or equal to 9.5 square metres) and compliant emergency travel distances (less than or equal to 30 metres) for rapid fire evacuation.

Financial Transparency & Democratized Real Estate: Instant access to itemized material costs calculated in localized currency (₹ INR) prevents cost inflation, protects individual home builders from opaque contractor estimates, and fosters trust across real estate transactions.

Inclusive Architectural Accessibility: The platform's lightweight, browser-based delivery ensures that users in low-bandwidth or resource-constrained environments can view and interact with modern 3D BIM models on ordinary laptops or mobile devices without specialized hardware.

7.3 Environmental Impact

The platform integrates computational environmental engineering directly into early-stage building design to reduce lifetime operational and embodied carbon:

Passive Solar Optimization (Heliodon Simulation): By providing a real-time diurnal sun dial 06:00 to 18:00, architects can simulate directional sunlight and cast accurate shadow angles across rooms. Positioning living zones and fenestrations according to solar paths maximizes natural daylighting, directly curbing reliance on artificial daytime electric lighting.

Natural Cross-Ventilation Verification: Auditing the window-to-floor surface area ratio 10 under green building norms ensures that spaces possess adequate natural ventilation conduits, mitigating internal thermal buildup and reducing recurring HVAC air-conditioning energy demands.

Embodied Carbon & Material Waste Reduction: Miscalculated material takeoffs lead to over-ordering of

concrete, bricks, and mortar, which contributes heavily to site scrap and landfill dumping. Deriving geometric material volumes (cubic metres of masonry, square metres of plaster and tile) directly from vector boundaries minimizes construction site material wastage.
Paperless Digital Workflows: Replacing multi-sheet physical blueprint printing and manual scale audits with a digital WebGL inspection canvas saves extensive paper consumption across design iteration cycles.

7.4 SDG Mapping

The objectives, methodologies, and outcomes of Patrick Architect map directly to the United Nations Sustainable Development Goals (UN SDGs):

Example

Goal Indicator	**Alignment Description**
☑ SDG 3: Good Health and Well-Being	Enforces minimum habitable volumes, adequate natural ventilation ratios, and statutory daylight standards, fostering healthy, mould-free indoor environmental quality (IEQ) for occupants.
☑ SDG 7: Affordable and Clean Energy	Incorporates a diurnal Heliodon sun-path study to maximize passive solar heat gains in winter and minimize cooling loads in summer, promoting clean passive energy efficiency in residential design.
☑ SDG 9: Industry, Innovation and Infrastructure	Fosters technological innovation by applying lightweight Computer Vision (OpenCV) and WebGL (Three.js) to modernize traditional infrastructure design pipelines without requiring high-cost hardware.
☑ SDG 11: Sustainable Cities and Communities	Supports safe and sustainable urban human settlements by automating National Building Code (NBC) safety audits for fire egress travel limits and structural clearances.
☑ SDG 12: Responsible Consumption and Production	Provides accurate, volume-derived Bill of Quantities (BOQ) takeoffs that eliminate over-estimation and construction material waste during brickwork, plastering, and flooring phases.
☑ SDG 13: Climate Action	Promotes low-carbon building design by optimizing natural lighting and ventilation conduits, directly reducing operational greenhouse gas emissions linked to residential HVAC and artificial illumination.

CHAPTER 8: CONCLUSION

Summarize

- Problem

  The Architecture, Engineering, and Construction (AEC) sector continues to face substantial workflow friction due to its reliance on flat, two-dimensional (2D) raster drawings and manual drafting practices. Non-technical clients and property buyers struggle to conceptualize vertical heights, volumetric bounds, and spatial circulation from static lines on paper. Concurrently, professional practitioners face steep labor overheads: converting 2D drawings into 3D models typically requires hours of repetitive manual tracing in expensive, proprietary CAD/BIM software. Furthermore, essential engineering inferences—such as statutory building code compliance audits (NBC/IBC regulations) and construction Bill of Quantities (BOQ) cost estimations—are handled sequentially through detached spreadsheets and physical scale rules, introducing human error, delaying project sanctions, and obscuring financial transparency during early design phases.

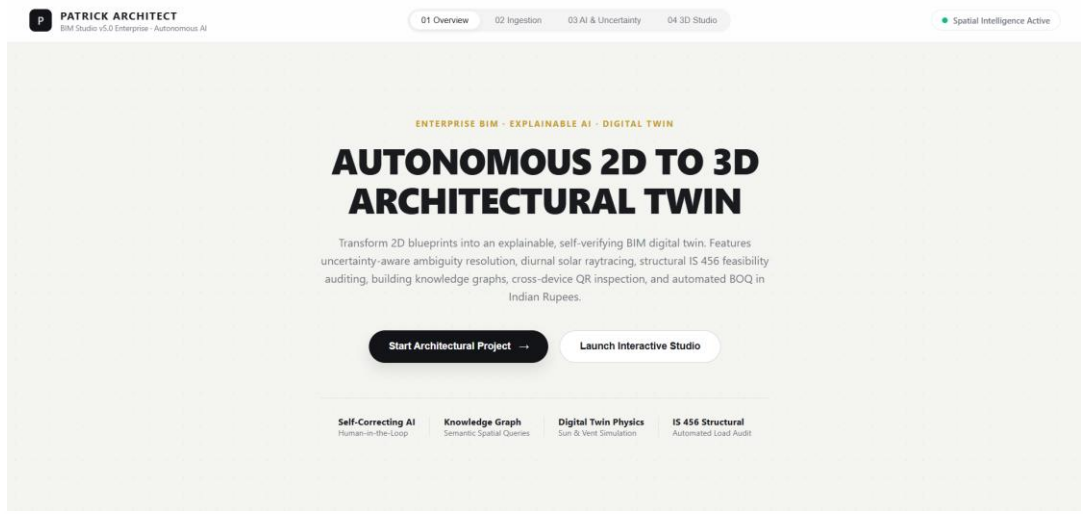
- Proposed solution

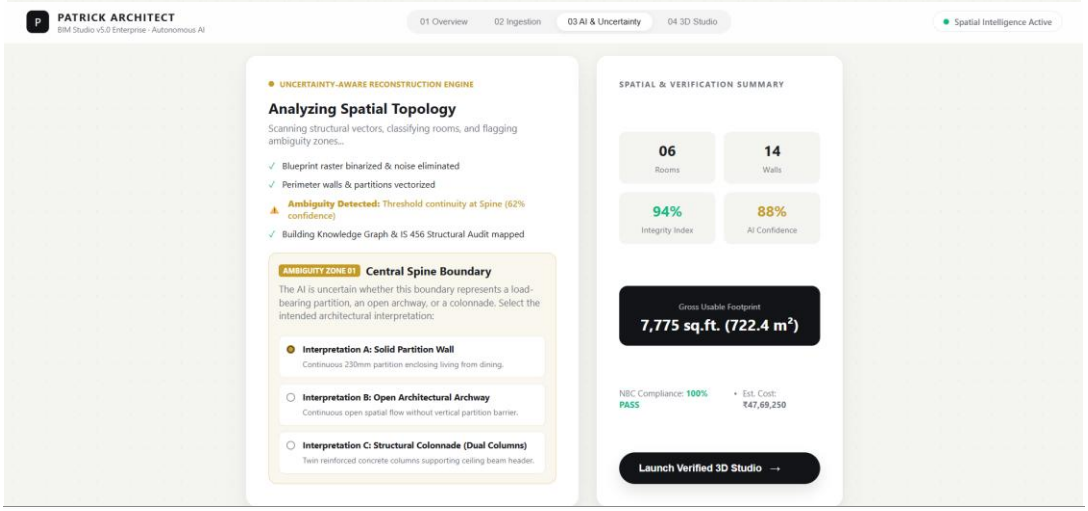
  To overcome these limitations, this project introduced Patrick Architect, an end-to-end, web-first computational architecture platform that bridges the divide between 2D drafting and interactive 3D Building Information Modeling (BIM). The system integrates a multi-tiered pipeline:
  Computer Vision Vectorization: Leverages OpenCV to binarize raster floor plans, eliminate text/annotation noise, isolate structural wall vectors via the Douglas-Peucker algorithm, and segment functional room cavities in under five seconds.
  Procedural 3D WebGL Studio: Procedurally extrudes calibrated metric geometry inside a Three.js canvas, supporting real-time architectural section plane slicing, diurnal heliodon solar studies ($06:00\text{ to }18:00$), and a 360-degree first-person walkthrough mode.
  Interactive Staging Canvas: Features an 18-piece architectural furniture and appliance catalog with real-time translation, multi-axis rotation, elevation adjustment, and uniform scaling directly across the floor plane.
  Automated BIM & Financial Inferences: Integrates a rule-based statutory engine verifying National Building Code of India (NBC 2016) parameters alongside an automated quantity takeoff engine estimating baseline material costs (brickwork, flooring, finishes, MEP) in Indian Rupees (₹ INR) based on CPWD Delhi Schedule of Rates (DSR).

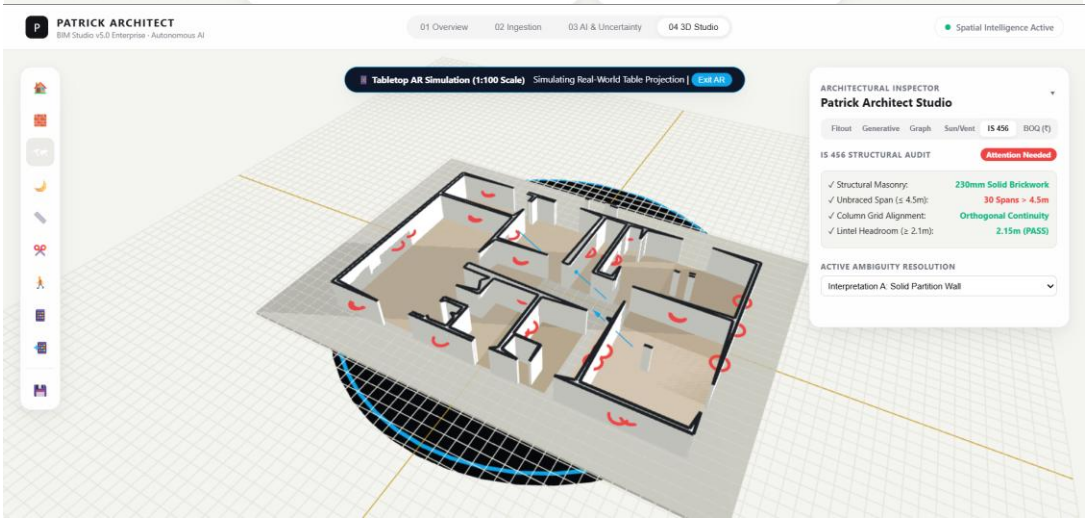
- Expected contribution

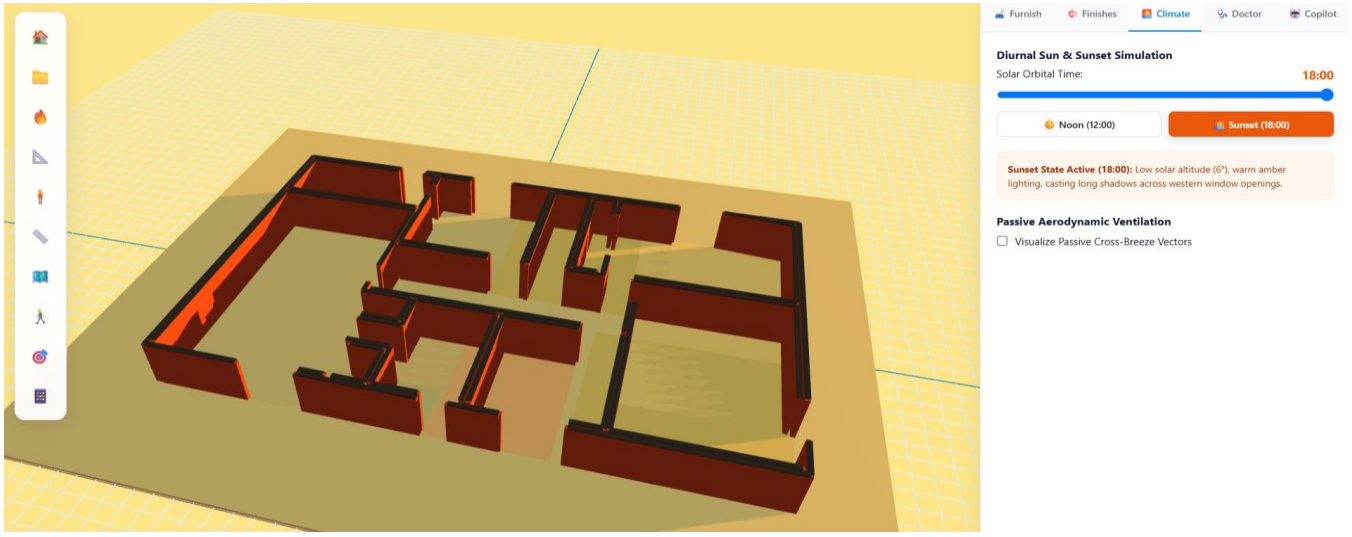
  The expected technical, industrial, and societal contributions of Patrick Architect are summarized as follows:
  Academic & Methodological Advancement: Extends standard 2D-to-3D computer vision research beyond raw boundary polygon extraction by coupling reconstructed geometry directly to functional BIM metadata, regulatory safety audits, and environmental solar simulations.
  Industrial Efficiency & Cost Democratization: Eliminates the necessity for expensive high-end modeling workstations and subscription-heavy proprietary software by deploying a lightweight, hardware-accelerated WebGL application natively accessible through any standard web browser at zero software cost.
  Standardized Regulatory Auditing: Automates preliminary statutory screening against municipal building codes, helping architectural firms and developers detect compliance violations (such as deficient room areas or excessive fire exit travel distances) before formal submission.

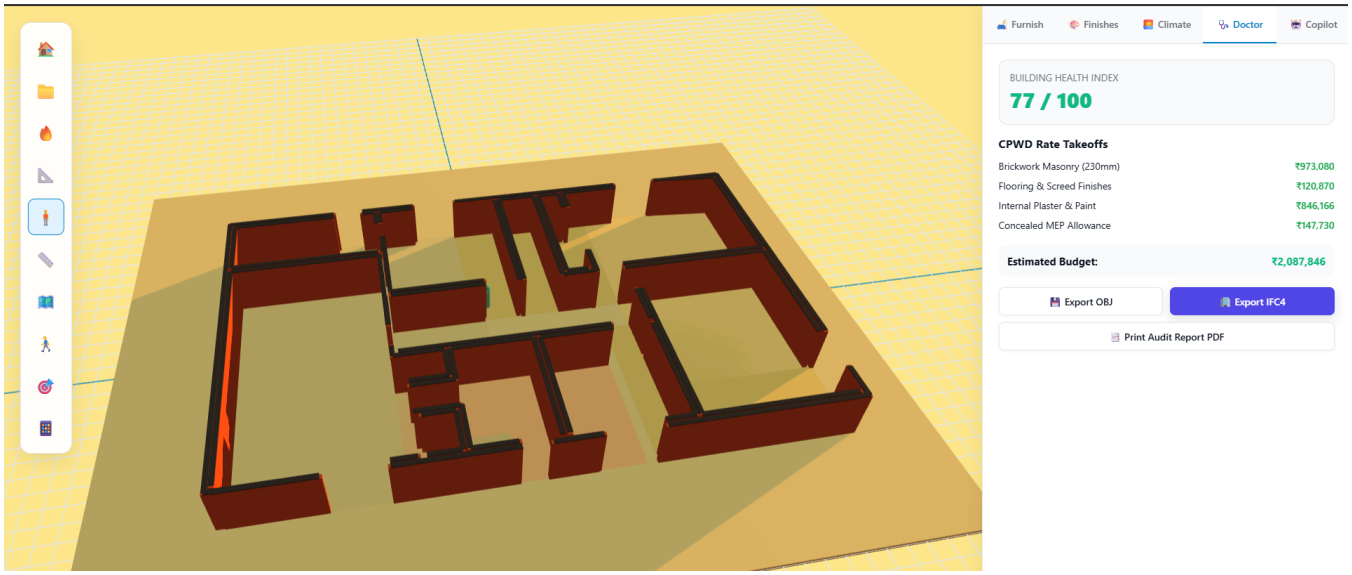
GUI


PATRICK ARCHITECT
BIM Studio v5.0 Enterprise - Autonomous AI
01 Overview
02 Ingestion
03 AI & Uncertainty
04 3D Studio
Spatial Intelligence Active
ENTERPRISE BIM · EXPLAINABLE AI · DIGITAL TWIN
AUTONOMOUS 2D TO 3D ARCHITECTURAL TWIN
Transform 2D blueprints into an explainable, self-verifying BIM digital twin. Features uncertainty-aware ambiguity resolution, diurnal solar raytracing, structural IS 456 feasibility auditing, building knowledge graphs, cross-device QR inspection, and automated BOQ in Indian Rupees.
Start Architectural Project →
Launch Interactive Studio
Self-Correcting AI
Human-in-the-Loop
Knowledge Graph
Semantic Spatial Queries
Digital Twin Physics
Sun & Vent Simulation
IS 456 Structural
Automated Load Audit


PATRICK ARCHITECT
BIM Studio v5.0 Enterprise - Autonomous AI
01 Overview
02 Ingestion
03 AI & Uncertainty
04 3D Studio
Spatial Intelligence Active
UNCERTAINTY-AWARE RECONSTRUCTION ENGINE
Analyzing Spatial Topology
Scanning structural vectors, classifying rooms, and flagging ambiguity zones...
Blueprint raster binarized & noise eliminated
Perimeter walls & partitions vectorized
Ambiguity Detected: Threshold continuity at Spine (62% confidence)
Building Knowledge Graph & IS 456 Structural Audit mapped
AMBIGUITY ZONE 01
Central Spine Boundary
The AI is uncertain whether this boundary represents a load-bearing partition, an open archway, or a colonnade. Select the intended architectural interpretation:
Interpretation A: Solid Partition Wall
Continuous 230mm partition enclosing living from dining.
Interpretation B: Open Architectural Archway
Continuous open spatial flow without vertical partition barrier.
Interpretation C: Structural Colonnade (Dual Columns)
Twin reinforced concrete columns supporting ceiling beam header.
SPATIAL & VERIFICATION SUMMARY
06
Rooms
14
Walls
94%
Integrity Index
88%
AI Confidence
Gross Usable Footprint
7,775 sq.ft. (722.4 m²)
NBC Compliance: 100%
PASS
Est. Cost:
₹47,69,250
Launch Verified 3D Studio →


PATRICK ARCHITECT
BIM Studio v5.0 Enterprise - Autonomous AI
01 Overview
02 Ingestion
03 AI & Uncertainty
04 3D Studio
Spatial Intelligence Active
Tabletop AR Simulation (1:100 Scale)
Simulating Real-World Table Projection |
Exit AR
ARCHITECTURAL INSPECTOR
Patrick Architect Studio
Fitout
Generative
Graph
Sun/Vent
IS 456
BOQ (₹)
IS 456 STRUCTURAL AUDIT
Attention Needed
✓ Structural Masonry:
230mm Solid Brickwork
✓ Unbraced Span (≤ 4.5m):
30 Spans > 4.5m
✓ Column Grid Alignment:
Orthogonal Continuity
✓ Lintel Headroom (≥ 2.1m):
2.15m (PASS)
ACTIVE AMBIGUITY RESOLUTION
Interpretation A: Solid Partition Wall


Furnish
Finishes
Climate
Doctor
Copilot
Diurnal Sun & Sunset Simulation
Solar Orbital Time:
18:00
Noon (12:00)
Sunset (18:00)
Sunset State Active (18:00): Low solar altitude (6°), warm amber lighting, casting long shadows across western window openings.
Passive Aerodynamic Ventilation
Visualize Passive Cross-Breeze Vectors


Furnish
Finishes
Climate
Doctor
Copilot
BUILDING HEALTH INDEX
77 / 100
CPWD Rate Takeoffs
Brickwork Masonry (230mm) ₹973,080
Flooring & Screed Finishes ₹120,870
Internal Plaster & Paint ₹846,166
Concealed MEP Allowance ₹147,730
Estimated Budget: ₹2,087,846
Export OBJ
Export IFC4
Print Audit Report PDF


Mobile Touch Inspection
Connect phone to same Wi-Fi and scan:
http://192.168.0.100:5000/mobile_view/87cd05bd
Close
Furnish
Finishes
Climate
Doctor
Copilot
BUILDING HEALTH INDEX
77 / 100
CPWD Rate Takeoffs
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Export IFC4
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